PV101 Day 4 Daily Run Sheet

Instructor-facing plan for Rich

CTS Cohort 6 - PV101 | Day 4 | Thursday, July 9, 2026

Topic: Small Solar System Build: PV Panel, Charge Controller, 12.8V LiFePO4 Battery, Wiring, and Inverter

Class setup: 6 students; 5 kits. 5 kits for 6 students: run 5 build stations. One station is a two-student team with Builder/Inspector roles that swap halfway. Solo builders still require a neighbor peer-check before energizing.

Teaching frame

- Main message: build the small solar system slowly enough that every part can be traced and explained.
- The real lab is not just making the inverter turn on. The real lab is safe sequence, polarity, protection, and documentation.
- 5 kits for 6 students: run 5 build stations. One station is a two-student team with Builder/Inspector roles that swap halfway. Solo builders still require a neighbor peer-check before energizing.
- Safety boundary: no energized wiring changes. Disconnect before touching conductors.

Learning targets

- Identify PV panel, charge controller, 12.8V LiFePO4 battery, inverter, protection, and conductors.
- Wire a small solar system in the correct general sequence: battery/controller first, PV second, inverter battery-side through protection.
- Use a multimeter to verify battery voltage, PV polarity/voltage, and controller-side readings before energizing.
- Explain why the inverter is protected and why it is not treated like a tiny controller load.
- Shutdown and document the system safely.

Materials

- 5 small solar system kits for 6 students
- Each kit: 1 PV panel, 1 charge controller, 1 12.8V LiFePO4 battery, 50Ah, 1 inverter, wiring/connectors, and required protection/disconnects
- Multimeters with leads
- Wire strippers
- Ratcheting crimpers and correct dies
- Appropriate ring terminals/lugs/ferrules/connectors
- Inline fuse holders, fuses, or demo breakers sized for the training kit
- Tape/labels/markers
- Safety glasses
- Panel covers, cardboard, towel, or face-down placement to keep PV inactive during wiring
- Small approved AC test load if inverter testing is allowed
- Student notebooks and measurement log sheets

Daily schedule

Time	Block	Instructor notes
10:00-10:10	Arrival + setup	Students get notebooks and quiz sheet. Instructor confirms kits remain disconnected: PV covered/unplugged, battery fuse out, inverter off.
10:10-10:20	Safety reset + goal	Frame the day: each group will build a small solar system, but nothing is energized until it is traced, metered, and approved.
10:20-10:35	Morning quiz	Individual, quiet. Use as a teaching check, not a trap. Humanity already invented enough traps, naturally.
10:35-10:55	Quiz review	Correct together. Tie missed answers back to today's physical build.
10:55-11:20	Slides part 1	System pieces: PV panel, charge controller, LiFePO4 battery, inverter, fuses/disconnects, conductors.
11:20-11:45	Instructor demo	Show correct connection order: battery to controller first, then PV to controller, then inverter to battery side through protection.
11:45-12:00	Whiteboard challenge	Students trace the path: sunlight -> PV -> controller -> battery -> inverter -> AC load. Mark every positive and negative path.
12:00-12:30	Lunch / reset	All kits remain disconnected. Batteries stored safely. Panels covered or face-down.
12:30-12:45	Kit assignment + roles	Set 5 stations for 6 students. One pair shares a kit and swaps Builder/Inspector roles. Explain peer-check standard.
12:45-1:05	Mechanical prep	Students mount/lay out controller, battery, inverter, protection, and wires. Strip/crimp as needed. No power yet.
1:05-1:35	Build step 1: battery side	Wire battery to charge controller battery terminals through correct polarity. Fuse remains removed/open until approval.
1:35-2:00	Build step 2: PV side	Wire PV panel to charge controller PV input. Panel stays covered/disconnected until meter checks and instructor approval.
2:00-2:20	Build step 3: inverter side	Wire inverter to battery side through fuse/disconnect. Do not

Time	Block	Instructor notes
		wire inverter output into anything permanent. Test only with approved small load.
2:20-2:40	Commissioning + measurements	Groups trace aloud, meter battery voltage (~12.8V nominal), verify PV polarity/voltage, then energize in approved order.
2:40-2:55	Cleanup + exit ticket	Disconnect PV first, turn inverter off, remove/open battery protection, document measurements, return kits.
2:55-3:00	Dismissal	Tools counted, scraps contained, batteries and panels safe. No mystery wires left behind, because we are not summoning demons.

Safety stop conditions

- Polarity is uncertain or readings do not match expectation.
- PV panel is producing voltage while students are still changing wiring.
- Battery fuse blows, breaker trips, arc/spark appears, or equipment smells hot.
- A conductor is warm, damaged, loose, or has exposed copper outside a terminal.
- A student wants to change wiring while anything is energized.
- Inverter wiring path is unclear or routed through the wrong controller terminals.
- The group cannot explain the full energy path clearly.

Whiteboard prompts

- Where does energy enter the system?
- Where is energy stored?
- Where is energy converted to AC?
- What gets connected first to most charge controllers?
- Where should protection be placed so it protects the most conductor?
- What measurement proves the system is ready for the next connection?

Exit ticket

1. Trace the system path in words: PV panel -> charge controller -> battery -> inverter -> load.
2. What gets connected first on most charge controllers: battery or PV? Why?
3. Name one STOP condition from today's lab.

Instructor notes after class

What landed:

_____

_____

What confused students:

_____

_____

Lab issues / tool needs:

_____

_____

Carry-forward for Day 5:

_____

_____

PV101 Day 4 Morning Quiz

PV wiring, polarity, protection, and small-system sequencing

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Topic: Small Solar System Build: PV Panel, Charge Controller, 12.8V LiFePO4 Battery, Wiring, and Inverter

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Name: _____ Date: _____

Directions: Answer in plain field language. Complete sentences are welcome; correct thinking matters more than fancy vocabulary.

1. 1. What does voltage describe in a circuit?

_____

_____

2. 2. What does current describe in a circuit?

_____

_____

3. 3. In a PV circuit, why do we verify polarity before connecting equipment?

_____

_____

4. 4. What usually happens to voltage when two matching modules are wired in series?

_____

_____

5. 5. What usually happens to current when two matching modules are wired in parallel?

_____

_____

6. 6. Why should a fuse or breaker be close to the source of energy?

_____

_____

7. 7. Why does a charge controller need correct battery connection and polarity?

_____

_____

8. 8. Why does an inverter usually connect to the battery side instead of the small controller load terminals?

_____

_____

9. 9. What should be checked before uncovering or connecting the PV panel?

_____

_____

10. 10. Before energizing the small solar system, what should the group do?

_____

_____

PV101 Day 4 Morning Quiz - Answer Key

PV wiring, polarity, protection, and small-system sequencing

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Class setup: 6 students; 5 kits. 5 kits for 6 students: run 5 build stations. One station is a two-student team with Builder/Inspector roles that swap halfway. Solo builders still require a neighbor peer-check before energizing.

Name: _____ Date: _____

Directions: Answer in plain field language. Complete sentences are welcome; correct thinking matters more than fancy vocabulary.

1. 1. What does voltage describe in a circuit?

Answer: Electrical pressure or potential difference.

2. 2. What does current describe in a circuit?

Answer: The flow of electric charge, measured in amps.

3. 3. In a PV circuit, why do we verify polarity before connecting equipment?

Answer: Reversed polarity can damage equipment and create unsafe conditions.

4. 4. What usually happens to voltage when two matching modules are wired in series?

Answer: Voltage adds; current stays about the same.

5. 5. What usually happens to current when two matching modules are wired in parallel?

Answer: Current adds; voltage stays about the same.

6. 6. Why should a fuse or breaker be close to the source of energy?

Answer: To protect conductors/equipment from overcurrent and reduce fault risk.

7. 7. Why does a charge controller need correct battery connection and polarity?

Answer: The controller must identify the battery/system voltage and reversed polarity can damage equipment.

8. 8. Why does an inverter usually connect to the battery side instead of the small controller load terminals?

Answer: An inverter can draw high current; it needs a properly protected battery connection unless the manufacturer says otherwise.

9. 9. What should be checked before uncovering or connecting the PV panel?

Answer: Circuit path, polarity, expected voltage, and instructor approval.

10. 10. Before energizing the small solar system, what should the group do?

Answer: Trace the circuit, verify polarity/voltage with a meter, check protection, and get instructor approval.

PV101 Day 4 Afternoon Lab

Build a small solar system from five training kits

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Class setup: 6 students; 5 kits. 5 kits for 6 students: run 5 build stations. One station is a two-student team with Builder/Inspector roles that swap halfway. Solo builders still require a neighbor peer-check before energizing.

Lab mission

- Build one small solar system per kit: 1 PV panel, 1 charge controller, 1 12.8V LiFePO4 battery, wiring, and 1 inverter.
- There are 5 kits and 6 students. One kit is shared by two students. That pair uses Builder/Inspector roles and swaps halfway.
- Success is not speed. Success is a system that can be traced, measured, protected, energized, shut down, and explained.

Required safety rules

- Polarity is uncertain or readings do not match expectation.
- PV panel is producing voltage while students are still changing wiring.
- Battery fuse blows, breaker trips, arc/spark appears, or equipment smells hot.
- A conductor is warm, damaged, loose, or has exposed copper outside a terminal.
- A student wants to change wiring while anything is energized.
- Inverter wiring path is unclear or routed through the wrong controller terminals.
- The group cannot explain the full energy path clearly.

Build sequence

1. **0. Assign stations:** 5 kits for 6 students: run 5 build stations. One station is a two-student team with Builder/Inspector roles that swap halfway. Solo builders still require a neighbor peer-check before energizing.
2. **1. Make it safe:** PV panel covered/face-down or unplugged. Battery fuse removed/open. Inverter off. Safety glasses on.
3. **2. Lay out the system:** Place PV panel, charge controller, LiFePO4 battery, protection, inverter, and wires so every connection can be seen and traced.
4. **3. Identify terminals:** Find and label controller BAT +/-, controller PV +/-, battery +/-, inverter +/-. Nobody trusts color alone, because colors have betrayed humanity repeatedly.
5. **4. Prep wires:** Cut/route wires with strain relief and service slack. Strip cleanly. Crimp terminals. Tug test every crimp.
6. **5. Wire battery to controller:** Connect battery side to controller BAT terminals with correct polarity. Keep fuse removed/open until instructor approval. Battery connects before PV for most controllers.

7. **6. Meter battery side:** Measure battery voltage directly, then at controller battery terminals. Expected battery is about 12.8V nominal for LiFePO4, but actual reading may vary.
8. **7. Wire PV to controller:** Connect PV leads to controller PV input with correct polarity. Keep panel covered/disconnected until ready to test.
9. **8. Meter PV side:** Before final PV connection or uncovering, verify PV open-circuit voltage and polarity. Call readings out loud.
10. **9. Wire inverter to battery side:** Wire inverter input to battery side through fuse/disconnect. Do not feed inverter from controller load terminals unless the equipment manual explicitly allows it.
11. **10. Peer check:** Neighbor or paired inspector traces full system and checks polarity, protection, crimp quality, and loose strands.
12. **11. Instructor approval:** Group explains circuit path and readings. Instructor approves before energizing.
13. **12. Commission:** Insert/close battery protection first, controller wakes up, uncover/connect PV, then turn inverter on. Test only with approved small load.
14. **13. Shutdown:** Turn off inverter, cover/disconnect PV, open/remove battery protection, then clean up and document results.

Measurement log

Check point	Expected / question	Measured value	Instructor initials
Battery direct	About 12.8V nominal LiFePO4; actual may vary		
Controller BAT terminals	Same polarity as battery		
PV panel open circuit	Polarity and voltage verified before connection		
Controller PV input	Correct polarity		
Inverter input	Battery-side protected DC input		
AC output test load	Only with approved small load		

Peer check before energizing

☐ Every + and - connection can be pointed to and named.

☐ Battery fuse/breaker is open or removed until final approval.

☐ PV panel is covered/face-down or disconnected until PV check.

☐ No loose strands, nicked conductors, or exposed copper outside terminals.

☐ Inverter input is battery-side through protection, not casually shoved into controller load terminals.

☐ Group can explain shutdown order.

Shutdown order

- Turn inverter off and remove AC load.
- Cover or disconnect PV panel.

- Open/remove battery protection.
- Verify controller and inverter are no longer energized as needed.
- Clean kit and return all parts.

Reflection

What mistake did your group catch before energizing?

_____

_____

What measurement mattered most?

_____

_____

What would you label better next time?

_____

_____